

for other diseases, for example. Nanotechnology could provide new tools for gathering detailed information about variations in disease states and about unique paths of treatment. Perhaps more important, nanotechnology could spur the personalized-medicine revolution by helping bring about real-time, sensitive monitoring of drug therapies. With more frequent feedback, a doctor could easily adjust drugs and dosages to personalize a patient's therapy. Indeed, treatments are expected to become more complex in the future. For instance, a doctor might prescribe a cocktail of several different drugs in calculated proportions—perhaps 10 percent of drug A, 50 percent of drug B and 40 percent of drug C. Nano-based monitoring devices could give doctors the ability to adjust the drug cocktail to suit the individual patient.

- 4. Participatory (regenerative) medicine** – The focus of this vision is to regenerate a part of the body that has been injured or has deteriorated as the result of disease, genetic defects or normal aging. Although regenerative medicine is generally associated with stem cells, nanotechnology could also lead to radically new treatments for spinal cord injury, macular degeneration, type 1 diabetes and other dysfunctions. Whether stem cells can be coaxed to rebuild tissues and restore function remains to be seen, but an equally promising approach is to employ nanostructured artificial tissues. It is still early, but many laboratories are experimenting with a wide variety of nano-material scaffolds that can be infused with cells to form artificial tissues, such as bone and liver. It appears possible to repair damaged nerves by injecting them with nanomaterials that form bridge-like lattices. Other nanostructures



Fountain of youth?

show promise as foundations for growing three-dimensional networks of blood vessels. Regenerating damaged tissues is one approach, but lost function might also be restored using nano-enhanced replacement parts for the body—devices that hook right into the nervous system. Although science fiction writers and moviemakers have

explored this idea in many colorful ways, it could indeed become reality as the result of rapid advances in nanotechnology, microelectronics, robotics, information science and neuroscience.

So it is quite fair to say that the sky seems to be the limit of what might one day be accomplished with nanostructured artificial tissues and nano-enhanced prosthetic devices. In the meantime, researchers developing them will have to